

Stablecoin



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Definition

In the narrowest sense, a *stablecoin* is a cryptocurrency that has its price pegged to a fiat currency (e.g., **USD**) and is fully backed by reserves denominated in the same fiat currency. More broadly, a *stablecoin* can be defined as a digital asset that has mechanisms to maintain a low deviation of its price from a target price.

Background

Since the invention of bitcoin and other cryptocurrencies, a major obstacle for their wider adoption and recognition as proper currencies, particularly as means of exchange and units of account, has been the instability (volatility) of their price in relation to fiat currencies. Stablecoins aim to overcome this obstacle.

Stablecoins also provide conveniences for cryptocurrency traders. They can sell cryptocurrency for stablecoins instead of fiat currency. And they can leverage their long positions in cryptocurrencies by taking loans in stablecoins collateralized by their cryptocurrency

holdings and selling the borrowed stablecoins for more cryptocurrency.

The most prominent stablecoin by market cap, Tether's **USDT**, was launched in 2014. The most prominent algorithmic stablecoin, **DAI** (Maker Foundation 2020), was launched in 2017. The number of stablecoin projects rose sharply since 2018 and, as of July 2022, stablecoins account for more than 15% of the total cryptocurrency market cap, 3 out of 10 top cryptocurrencies are stablecoins, and the daily trading volume of **USDT** is larger than the volumes of **BTC** and **ETH** combined.

The rising demand for stablecoins also led to the launch of cryptocurrencies that claimed to be stablecoins but that later proved to be unstable. Most notably, the collapse of Terra (**UST** and **LUNA**) in May 2022 had a cascading effect that wiped approximately 50% of the total market cap of the entire cryptocurrency industry by the end of that year.

Theory and Application

Various mechanisms can contribute to a stablecoin's stability. Ultimately, they are all grounded on the basic economic principles of supply and demand. If demand for buying/selling stablecoins is higher than the current supply of sale/purchase orders, this supply must be increased to avoid an increase/decrease in the stablecoin's price.

Backing the stablecoins by reserves and using these reserves to actively buy and sell stablecoins for prices close to the target price is currently the most common mechanism. This mechanism is not exclusive to stablecoins; it can be seen in pegged national currencies such as the **HKD** (Hong Kong Dollar). Centrally operated fiat-pegged fiat-backed stablecoins such as **USDT** and **USDC** use variations of this mechanism that may differ, for instance, on the actual composition of the reserves and with whom the operator interacts to buy and sell stablecoins. As long as the stablecoin is fully backed by reserves in the currency to which it is pegged and the operator can react quickly enough to variations in the demand, it is easy to see that stability will be guaranteed. Typically, the reserves will not be kept all in cash in a vault, safe, or bank account, but rather partly in interest-bearing financial instruments such as bonds. The returns from such investments provide revenue for the operator. The risks associated with these investments may imply that the stablecoin may eventually lose its full-backing, compromising the stability in the long term. The lack of liquidity of these investments may cause the operator to be unable to react quickly enough to changes in demand, compromising the stability in the short term. Another source of revenue is the fees or spread practiced when buying and selling the stablecoin. For example, if the operator sells **USDT** for 1.005 **USD** and buys **USDT** for 0.995 **USD**, it has a revenue of 1 cent for every **USDT** that it buys and then sells, while keeping the price stable within the range between 0.995 and 1.005. The main drawback of fiat-backed stablecoins is that it requires trust on the entities keeping the reserves. This is not only a theoretical concern. For instance, the lack of transparency about the reserves and skepticism about its full-backing claim, combined with inefficient stabilization measures by Tether, have actually caused **USDT** to trade for as low as 0.91 **USD**.

Interestingly, issues related to transparency of the reserves do not arise when the backing asset is a cryptocurrency on a public blockchain. For example, Wrapped BTC (**WBTC**) is a stable-

coin issued on the Ethereum blockchain, pegged to **BTC**, and backed by **BTC** held on the Bitcoin blockchain. Because Bitcoin is a public blockchain, anyone can see how much **BTC** is available in the reserves backing **WBTC**. Such *wrapping* can be considered a major application of stablecoins, since it enables interoperability between blockchains through *bridges*. In the particular case of **WBTC**, it enables the users to essentially use (an asset pegged to) **BTC** on Ethereum's contracts. Nevertheless, despite the transparency, there are still issues related to the custody of the backing assets. Bridges at the time of writing are one of the weakest points in blockchain security. They have been subject to many cyber-attacks that stole an estimated total of 2 Billion **USD** worth of backing assets up to August 2022, and such thefts amounted to an estimated 69% of the total funds stolen in 2022 (Chainalysis Team 2022).

The asset backing a stablecoin does not need to be the same asset to which the stablecoin is pegged. In particular, stablecoins pegged to **USD** and backed by cryptocurrencies are possible, as exemplified by Neutrino **USD** (**USDN**) (Ivanov and Pupyshev 2020), which is pegged to **USD**, issued on the waves blockchain, and backed by **WAVES**, and **SIGMAUSD** (Emurgo Team 2021) (based on the Djed stablecoin protocol (Zahmentferner et al. 2021, 2023)), which is pegged to **USD**, issued on the Ergo blockchain, and backed by **ERG**. When the backing asset's price falls, the stablecoin may cease to be fully backed. This is an inherent risk that cannot be completely avoided. Nevertheless, in order to reduce its probability, crypto-backed stablecoins typically aim for a generous surplus of reserves to cushion price falls.

Crypto-collateralized stablecoins, of which **DAI** is currently the most prominent example, are similar to crypto-backed stablecoins, with the subtle but important difference, however, that cryptocurrencies are used to provide, instead of reserves, collateral for stablecoin-denominated loans. Similarly to how a house that serves as collateral in a mortgage does not count as reserves in the bank that is providing the loan, the collateral provided for the issuance of a

crypto-collateralized stablecoin does not count as reserve. Due to the lack of reserves, pure crypto-collateralized stablecoins resemble an extreme form of fractional reserve banking (with a fraction of 0%). Unlike commercial bank money (i.e., money in a bank account), which can be exchanged at any time by anyone holding it for central bank money by withdrawing cash from the bank's reserves, crypto-collateralized stablecoins cannot be exchanged for anything by withdrawing from the stablecoin contract's reserves, simply because there are no reserves. When interacting with the contract, the users can only use stablecoins to repay their own loans and recover their own collateral or to bid in auctions for liquidated collateral from others.

Although backing and collateralization facilitate stability and tend to increase the public's trust on the stability, it is not impossible to achieve stability without backing or collateralization. To see this, consider that a central bank digital currency (CBDC) would be pegged to the central bank's fiat currency and, like the fiat currency itself, would not need to be backed by anything, by definition of *fiat*. And, despite this, it would nevertheless be stable in relation to the prices of goods and services, at least in the short term, just like their associated fiat currencies in countries that enjoy low inflation. Central banks achieve stability of their fiat currencies (and of their CBDCs should they deploy them) by adjusting the circulating supply. For instance, they may print more money to increase the supply, and they may increase the base interest rate to encourage people and financial institutions to buy bonds, thereby temporarily taking money out of circulation. Unbacked stablecoins may implement similar stabilization mechanisms. One of the earliest examples is *seigniorage shares* (Sams 2014). To take stablecoins out of circulation, the users are encouraged to sell their stablecoins in exchange for seigniorage shares; conversely, when the supply needs to be increased, stablecoins are issued and distributed to seigniorage shareholders. While this mechanism may seem convincing at first sight, the collapse of Terra (UST and LUNA), which can be considered a variation of this mechanism, demonstrated that this mechanism is not without issues.

In addition to mimicking traditional stabilization mechanisms, unbacked stablecoins can pursue novel stabilization mechanisms. In contrast to physical cash, which is difficult to track and destroy after being issued, a purely digital asset can be easily destroyed to reduce the supply. For example, Ampleforth (Kuo et al. 2019) simply rebases the balances of all users. If the price of 1 AMPL is currently 0.5 USD and the target is for it to be 1 USD, then all balances are smoothly halved. A user that had 10 AMPL worth 5 USD will eventually have 5 AMPL worth 5 USD. This is done fairly, automatically, and efficiently to all users and not only to people who can access financial services.

In contrast to traditional cryptocurrencies, which are tied to the blockchains to which they are native, it is common for a single stablecoin to exist on many different blockchains. For example, USDT has already been issued in at least seven blockchains. But, while the trend of issuing stablecoins as assets on the existing general-purpose blockchains (e.g., ERC-20 contracts on Ethereum) is dominant, there were also cases of blockchains developed specifically for the purpose of hosting stablecoins (cf. Terra (Kereiakes et al. 2019) and Libra blockchain (Amsden et al. 2020)).

Furthermore, while fiat-pegged stablecoins are dominant by market cap, stablecoins that are not pegged to fiat currencies are starting to appear. RAI (Inoescu and Soleimani 2020), for instance, is a variant of DAI that uses a PID controller to dynamically adjust the target price.

With all types of stablecoins, various degrees of automation of the stabilization mechanisms, both on-chain and off-chain, are possible. Stablecoins with a high degree of automation, especially with on-chain contracts, are often called *algorithmic*. Because such stablecoins execute and profit mostly autonomously, they are usually associated with a second digital asset that gives its holders the right to benefit from the execution of the contract. For example, DAI has MKR, seigniorage share stablecoins have shares, and Djed has ReserveCoins. Often, these assets also give governance responsibilities and rights to their holders, to set the parameters of the algorithms.

Open Problems and Future Directions

The legal status of stablecoins and the requirements for stablecoin issuers and operators are unclear and undefined at the moment and remain as a regulatory open problem. This uncertainty is believed to have been the primary cause for the decision to abort the launching of major stablecoin projects such as **DIEM**. Particularly, in the case of algorithmic stablecoins implemented as smart contracts, there seems to be room for regulatory innovation that would recognize their high degree of transparency and autonomous execution, which distinguishes their developers and issuers from traditional financial institutions.

On the technical side, many stablecoin projects are still vague about their stabilization mechanisms and their robustness. This may be a case of stability by obscurity similar to the “security by obscurity” that used to prevail before modern cryptography. The future work could devote more time to formal arguments, proofs, and stress testing simulations of stability.

Furthermore, the effect of crypto-backed and crypto-collateralized stablecoins on the stability of the assets used for backing or collateralization has not been sufficiently researched yet. Since these stablecoins allow leveraging long positions on those assets, it seems reasonable to expect that they may potentially lead to price bubbles and greater instability for those assets.

More generally, not much attention has been devoted to the stabilization of the primary native cryptocurrencies that power the blockchains (e.g., **BTC**, **ETH**). This may be because the public expects and hopes that the cryptocurrency prices will grow instead of remaining pegged. However, mechanisms that would allow growth with low volatility are plausible. For instance, more dynamic monetary policies could be used in order to adjust the supply more effectively; decentralized buy-back mechanisms could be implemented to dampen price crashes. If the relative market cap of stablecoins grows and new stablecoin-specific blockchains are launched,

traditional cryptocurrencies may feel forced to implement stabilization mechanisms to remain competitive and attractive to users.

In a distant future, if some cryptocurrencies ever achieve the initial dream of replacing traditional fiat currencies, pegging stablecoins to fiat currencies will become irrelevant. This suggests that unpegged stablecoins or at least stablecoins pegged to assets or indexes other than fiat currencies may be an important long-term direction for research and development. Furthermore, fiat currencies are unbacked for a reason: it is not economically efficient to keep full reserves of assets uselessly stored in a vault; it is better to use those assets for the benefit of society. It is even less efficient to keep reserves or collateral with large surpluses in the case of crypto-backed and crypto-collateralized stablecoins. Therefore, even though unbacked stablecoins have been a minority so far and have been significantly more volatile than backed stablecoins, it is plausible that unbacked (or only partially backed) stablecoins will become more prominent in the future. However, as the collapse of Terra in 2022 demonstrated, more research in this direction is needed.

Naturally, the risk of disruption of the traditional monetary order by cryptocurrencies, especially if they are stable or claiming to be stable and then proving not to be, has not gone unnoticed. This risk is one (though not the only) motivation for much of the ongoing work by national monetary authorities on CBDCs, which can be considered a special type of stablecoin according to the broader definition used here. The interoperability and interplay between CBDCs and stablecoins operated by private companies or executed autonomously by smart contracts will be interesting to observe as well and remains as another direction for future work.

Cross-References

- [Bitcoin](#)
- [Layer 2 Protocols](#)
- [Sidechains](#)

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